

1. **Stating the hypotheses.** In each of the following situations a significance test for a population mean μ is called for. Identify the null hypothesis H_0 and the alternative hypothesis H_A in each case. Make sure to use proper notation! (*e.g.* $H_0 : \mu = \mu_0$).

- (a) A company produces an item that is supposed to have a six inch hole punched in the center. A quality control inspector is concerned that the machine which punches the hole is “out-of-control”. In an effort to test this, the inspector is going to gather a random sample of items punched by the machine and measure the diameter of the holes.

$$H_0 : \mu = 6 \quad \text{versus} \quad H_a : \mu \neq 6$$

- (b) The managers at Hunter Chemical Company claim that their major product contains, on average, 4.0 fluid ounces of caustic material per gallon. They further state that the distribution of caustic material is normal with a standard deviation of 1.3 fluid ounces. If there is too much caustic material, the product will be dangerous. A government inspector believes that the product is indeed dangerous. She randomly selects a sample of 100, gallon-size containers of the product and finds that their mean weight of caustic materials is 4.5 fluid ounces per gallon.

$$H_0 : \mu = 4 \quad \text{versus} \quad H_a : \mu > 4$$

- (c) The University bookstore tells prospective students that the average cost of its textbooks is \$52 with a standard deviation of \$4.50. A group of statistics students thinks that the average cost is higher. In order to test the bookstore’s claim against their alternative, the students select a random sample of size 100. Assume that the mean from their random sample is \$52.80.

$$H_0 : \mu = 52 \quad \text{versus} \quad H_a : \mu > 52$$

- (d) The FMA Company has designed a new type of 16 lb bowling ball. They believe that this ball would improve a person’s bowling score. The company knows that the average man who bowls in a scratch league with the company’s old ball has a bowling score of 155. The variance of these scores is 100. The company asks a random sample of 100 men bowling in scratch leagues to bowl for five weeks with their new ball. The mean bowling score for these men with the new ball is 170. There is no reason to believe the variance is any different with the new ball.

$$H_0 : \mu = 155 \quad \text{versus} \quad H_a : \mu > 155$$

- (e) A school principal wants to test if it is true that high school juniors use the computer an average 3.2 hours a day. He selects a random sample of 50 high school juniors and finds out that they use the computer an average of 3.1 hours a day.

$$H_0 : \mu = 3.2 \quad \text{versus} \quad H_a : \mu \neq 3.2$$

2. **Diets.** Developers of diets like to advertise that their diets produce significant weight loss results. A new diet has been developed and several studies were done to investigate the effectiveness of the diet. In each study the average number of pounds lost was of interest, and researchers would like to show that dieters lost a significant amount of weight.

(a) Identify μ in this scenario. (**Choose One**)

- Average number of pounds lost by all people in the sample.
- True average weight of all people in the population.
- **True average number of pounds lost by all people in the population.**
- Average weight of all people in the sample.

(b) Identify the appropriate null and the alternative hypotheses to test the researchers' claim.

$$H_0 : \mu = 0 \quad \text{versus} \quad H_a : \mu > 0$$

(c) Suppose that an average weight loss of 1 pound ($\bar{x} = 1$) and standard deviation of $s = 7$ pounds were found in each of the following scenarios. Further, assume weight loss is normally distributed. For each case, compute the test statistic, find the approximate p -value and make a decision about the null hypothesis using an $\alpha = 0.05$ level of significance. Round t -statistics to two decimals.

i. **Scenario 1:** a sample of size $n = 10$

A.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1 - 0}{7/\sqrt{10}} = \mathbf{0.45} \quad df = \mathbf{9}$$

B. The p -value corresponding to the correct test statistic is 0.331. This p -value suggests **Little or no evidence** against the null hypothesis.

C. Decision:

- Reject the null hypothesis because the p -value $\leq \alpha$.
- Reject the null hypothesis because the p -value $> \alpha$.
- Fail to reject the null hypothesis because the p -value $\leq \alpha$.
- **Fail to reject the null hypothesis because the p -value $> \alpha$.**

ii. **Scenario 2:** a sample of size $n = 120$

A.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1 - 0}{7/\sqrt{120}} = \mathbf{1.56} \quad df = \mathbf{119}$$

B. The p -value corresponding to the correct test statistic is 0.0601. This p -value suggests **Moderately strong evidence** against the null hypothesis.

C. Decision:

- Reject the null hypothesis because the p -value $\leq \alpha$.
- Reject the null hypothesis because the p -value $> \alpha$.
- Fail to reject the null hypothesis because the p -value $\leq \alpha$.

- Fail to reject the null hypothesis because the $p\text{-value} > \alpha$.

iii. **Scenario 3:** a sample of size $n = 1050$

A.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1 - 0}{7/\sqrt{1050}} = 4.63 \quad df = 1049$$

B. The p -value corresponding to the correct test statistic is 0.000002. This p -value suggests Extremely strong evidence against the null hypothesis.

C. Decision:

- Reject the null hypothesis because the $p\text{-value} \leq \alpha$.
- Reject the null hypothesis because the $p\text{-value} > \alpha$.
- Fail to reject the null hypothesis because the $p\text{-value} \leq \alpha$.
- Fail to reject the null hypothesis because the $p\text{-value} > \alpha$.

D. A weight loss of one pound is found to be statistically significant. The researchers share their results with the public claiming that their diet produces meaningful, significant weight loss. Do you agree? Explain your answer.

From a practical standpoint, losing one pound is not a significant amount of weight.

3. [13pts.] **Sunflower seedlings.** A biologist was interested in determining whether sunflower seedlings treated with an extract from *Vinca minor* roots resulted in a lower average height of sunflower seedlings than the standard height of 15.7 cm. The biologist treated a random sample of 33 seedlings with the extract and subsequently measured the height of those seedlings. The data can be found in the file *Height.jmp* on Blackboard.

(a) Using SA, R, JMP, or Excel find the values of the mean and the standard deviation for these data. Round your answers to two decimal places.

$$\bar{x} = 15.41 \quad s = 0.42$$

(b) At the 0.01 level of significance, is there evidence that the true average height of the seedlings treated with an extract from *Vinca minor* roots is less than 15.7 cm?

i. Identify the null and alternative hypothesis, H_0 and H_a :

$$H_0 : \mu = 15.7 \quad \text{vs} \quad H_A : \mu < 15.7$$

ii. Report the test statistic. $t = -3.99$

iii. How many degrees of freedom are associated with your test statistic? $df = 32$

iv. Report the appropriate p -value. $p = 0.0002$

v. Identify the appropriate decision.

- Since the p -value is less than $\alpha = 0.01$, we reject the null hypothesis.

- Since the p -value is greater than $\alpha = 0.01$, we reject the null hypothesis.
 - Since the p -value is less than $\alpha = 0.01$, we fail to reject the null hypothesis.
 - Since the p -value is greater than $\alpha = 0.01$, we fail to reject the null hypothesis.
- vi. Based on your decision, write a conclusion in the context of the problem.

There is statistically significant evidence at the $\alpha = 0.01$ level to indicate that the sunflower seedlings treated with an extract from *Vinca minor* roots resulted in a lower average height of sunflower seedlings than the standard height of 15.7 cm.

- (c) Explain what a Type I Error is in the context of this problem.

A Type I error in this situation is concluding that the true average height of the seedlings treated with an extract from *Vinca minor* roots is less than 15.7 cm when in reality it is not.

4. Answer the following questions on-line by checking either **True** or **False**.

- (a) When the p -value is larger than α , we accept the null hypothesis. **False**
- (b) If the population mean is known, there is no need to test hypotheses concerning the population mean. **True**
- (c) Level of confidence is another name for level of significance. **False**
- (d) If we reject a null hypothesis at the 1% level, we also reject it at the 5% level. **True**
- (e) The decision to use a one-sided or two-sided test is usually made after the data is analyzed. **False**